Homework 02 Data Handling, Graphics, More R

Due by 11:59pm, Tuesday 2.3.26

S&DS 2300/5300/ENV 757

Submission Details

This homework assignment contains 1 long problem. Edit this `.Rmd` file and insert your responses under the appropriate problems. Submit the knitted output of this file either as a .pdf file.

Your document should look nicely formatted, with R code, plots, and text descriptions nicely integrated.

CRITICAL 1 - for every included chunk below, remove `eval = F` before knitting!!!!

CRITICAL 2 - Please submit knitted version of this assignment as a PDF file on GRADESCOPE You can knit directly to a DOC and then convert to PDF in Word (choose SAVE AS), or you can knit directly to PDF if that works for you.

DELETE LINES 19 THROUGH 29 (INCLUSIVE) BEFORE KNITTING AND SUBMITTING!

(1) OK Cupid Dataset This is a dataset first discussed by Kirkegaard and Bjerrekaer. We'll use this dataset several times this semester.

The data itself is stored as a parquet. This is a format where each column is stored separately (rather than as a giant array or matrix). It is more efficient to quickly access are search data. We will create a the usual dataframe.

Run the code below to get the dataset itself and then to get a dataset that is the list of variables in the dataset.

```
#Run the line below one time only
#install.packages("arrow")
#install.packages("tzdb")
library(arrow)
```

```
##
## Attaching package: 'arrow'

## The following object is masked from 'package:utils':
##
##     timestamp
```

```
#get the data - this takes a moment to load
okcupid <- data.frame(read_parquet("https://github.com/jreuning/sds230_data/raw/refs/heads/main/OKCupid/

#get the dataset of questions
okcupidQ <- read.csv("https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/OKCupid/que
```

1.1) Write a few lines of code to get a sense of both the dataset and the variables in contains. Include whatever code you use below. Be sure to calculate the size of the dataset. For all code you write, always include a comment line on what you are doing.

As a part of this, you may want to open `okcupidQ` in the Environment (upper right quadrant of RStudio, click on Environment, double click on the dataset.) and browse throught the questions. DO NOT INCLUDE OUTPUT AS A RESULT OF USING head() or tail().

Write code to get the unique values of `Type` in `okcupidQ`. What do you think these values mean?

Next, write not more than three sentences about what you observe about the dataset. Comment on missing values, the types of questions being asked, and whatever else you might notice.

```
# display head of both datasets in addition to opening the data in environment
#head(okcupid)
#head(okcupidQ)

# understand structure of both
str(okcupid)
```

```
## 'data.frame':    68371 obs. of  2626 variables:
##  $ Unnamed..0          : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ q2                  : chr  NA NA NA NA ...
##  $ q11                 : chr  "Horrified" NA NA NA ...
##  $ q12                 : chr  NA NA NA NA ...
##  $ q13                 : chr  NA NA "No" NA ...
##  $ q14                 : chr  NA NA "No" NA ...
##  $ q16                 : chr  NA NA NA NA ...
##  $ q17                 : chr  "No" NA "No" NA ...
##  $ q18                 : chr  NA NA NA NA ...
##  $ q20                 : chr  NA NA NA NA ...
##  $ q22                 : chr  NA NA NA NA ...
##  $ q24                 : chr  NA NA NA NA ...
##  $ q25                 : chr  NA NA NA NA ...
##  $ q26                 : chr  NA NA NA NA ...
##  $ q28                 : chr  "No" NA "No" NA ...
##  $ q29                 : chr  NA NA NA NA ...
##  $ q30                 : chr  NA NA NA NA ...
##  $ q32                 : chr  "I have little or no interest." NA "I have little or no interest." NA
##  $ q33                 : chr  "Yes" NA NA NA ...
##  $ q35                 : chr  "Love" NA "Love" NA ...
##  $ q36                 : chr  NA NA NA NA ...
##  $ q37                 : chr  NA NA NA NA ...
##  $ q38                 : chr  NA NA NA NA ...
##  $ q39                 : chr  NA NA "No" NA ...
##  $ q41                 : chr  NA NA "Extremely important" NA ...
##  $ q42                 : chr  NA NA NA NA ...
##  $ q43                 : chr  NA NA NA NA ...
##  $ q45                 : chr  NA NA NA NA ...
##  $ q46                 : chr  NA NA "Good" NA ...
##  $ q48                 : chr  NA NA NA NA ...
##  $ q49                 : chr  "Carefree" NA "Carefree" NA ...
##  $ q50                 : chr  NA NA NA NA ...
##  $ q55                 : chr  "No" NA "No" NA ...
##  $ q56                 : chr  "No" NA NA NA ...
```

```
##  $ q57                    : chr  NA NA NA NA ...
##  $ q60                    : chr  NA NA "Warm-hearted" NA ...
##  $ q61                    : chr  NA NA NA NA ...
##  $ q63                    : chr  NA NA NA NA ...
##  $ q65                    : chr  NA NA NA NA ...
##  $ q67                    : chr  NA NA NA NA ...
##  $ q68                    : chr  NA NA NA NA ...
##  $ q69                    : chr  NA NA NA NA ...
##  $ q70                    : chr  "No" NA "No" NA ...
##  $ q71                    : chr  "No" NA "No" NA ...
##  $ q73                    : chr  NA NA "Big city" NA ...
##  $ q74                    : chr  NA NA "Clothes" NA ...
##  $ q76                    : chr  NA NA NA NA ...
##  $ q77                    : chr  "Sometimes" NA "Rarely" "Sometimes" ...
##  $ q78                    : chr  NA NA NA NA ...
##  $ q79                    : chr  "Never." NA "I smoked in the past, but no longer." NA ...
##  $ q80                    : chr  "I never do drugs." NA NA NA ...
##  $ q81                    : chr  "Rarely" NA "Rarely" NA ...
##  $ q82                    : chr  NA NA NA NA ...
##  $ q84                    : chr  NA NA "No" NA ...
##  $ q86                    : chr  NA NA "No" NA ...
##  $ q87                    : chr  "No" NA "No" NA ...
##  $ q90                    : chr  NA NA NA NA ...
##  $ q91                    : chr  NA NA "Read" NA ...
##  $ q92                    : chr  NA NA NA NA ...
##  $ q95                    : chr  NA NA NA NA ...
##  $ q96                    : chr  NA NA NA NA ...
##  $ q97                    : chr  NA NA NA NA ...
##  $ q105                   : chr  "No" NA "No" NA ...
##  $ q106                   : chr  "No" NA "No" NA ...
##  $ q109                   : chr  NA NA NA NA ...
##  $ q110                   : chr  NA NA NA NA ...
##  $ q111                   : chr  NA NA NA NA ...
##  $ q112                   : chr  NA NA NA NA ...
##  $ q113                   : chr  NA NA NA NA ...
##  $ q114                   : chr  NA NA NA NA ...
##  $ q122                   : chr  "A lot" NA "A lot" NA ...
##  $ q123                   : chr  "No" NA "No" NA ...
##  $ q124                   : chr  NA NA NA NA ...
##  $ q125                   : chr  NA NA NA NA ...
##  $ q126                   : chr  NA NA NA NA ...
##  $ q127                   : chr  NA NA NA NA ...
##  $ q128                   : chr  "I have no tattoos" NA "I have no tattoos" NA ...
##  $ q129                   : chr  NA NA "No" NA ...
##  $ q130                   : chr  NA NA NA NA ...
##  $ q132                   : chr  NA NA NA NA ...
##  $ q133                   : chr  NA NA NA NA ...
##  $ q134                   : chr  NA NA NA NA ...
##  $ q135                   : chr  NA NA NA NA ...
##  $ q136                   : chr  "Yes" NA "Yes" NA ...
##  $ q137                   : chr  "No" NA NA NA ...
##  $ q140                   : chr  NA NA NA NA ...
##  $ q141                   : chr  NA NA "Yes" NA ...
##  $ q144                   : chr  NA NA NA NA ...
```

```
##  $ q146                 : chr  "Kissing in a tent, in the woods" NA "Kissing in Paris" NA ...
##  $ q152                 : chr  NA NA NA NA ...
##  $ q154                 : chr  NA NA NA NA ...
##  $ q157                 : chr  NA NA NA NA ...
##  $ q159                 : chr  NA NA NA NA ...
##  $ q160                 : chr  NA NA NA NA ...
##  $ q161                 : chr  NA NA "More freedom" NA ...
##  $ q163                 : chr  NA NA NA NA ...
##  $ q165                 : chr  NA NA NA NA ...
##  $ q166                 : chr  NA NA NA NA ...
##  $ q170                 : chr  NA NA NA NA ...
##   [list output truncated]
```

```r
str(okcupidQ)
```

```
## 'data.frame':    2620 obs. of  10 variables:
##  $ X       : chr  "q2" "q11" "q12" "q13" ...
##  $ text    : chr  "Breast implants?" "How does the idea of being slapped hard in the face during sex
##  $ option_1: chr  "more cool than pathetic" "Horrified" "Yes" "Yes" ...
##  $ option_2: chr  "more pathetic than cool" "Aroused" "No" "No" ...
##  $ option_3: chr  "" "Nostalgic" "" "" ...
##  $ option_4: chr  "" "Indifferent" "" "" ...
##  $ N       : int  24839 28860 22496 32581 31127 8341 38180 21686 16643 31024 ...
##  $ Type    : chr  "N" "N" "O" "O" ...
##  $ Order   : chr  "" "" "" "" ...
##  $ Keywords: chr  "sex/intimacy; preference; opinion" "sex/intimacy" "sex/intimacy" "sex/intimacy" .
```

```r
# get the dimensions of the original dataset - there are 68371 rows and 2626
# columns meaning 68371 responses on potentially 2626 questions
dim(okcupid)
```

```
## [1] 68371  2626
```

```r
# count n/as
missing_values <- colSums(is.na(okcupid))
sorted_missing_values <- sort(missing_values, decreasing = T)
head(sorted_missing_values, 10)
```

```
##   q92899  q72000 q155299 q125283 q145537 q116932 q128112  q91348 q118236  q91207
##    68270   68252   68251   68245   68223   68218   68213   68209   68206   68197
```

REPLACE THIS TEXT WITH YOUR RESPONSE AND MAKE SURE THE RESPONSE IS IN ITALICS Results were of the sexual nature

1.2) Write exactly three questions that you are curious to answer using this dataset.

What are most answered questions (what has the highest response percentage). What is the least answered question (what has the lowest response percentage).

1.3) We are going to calculate a numeric value of income. I did this in the code below.

```r
grep("income", names(okcupid)) # this looks the word income in the column names of okcupid; this allows
```

```
## [1] 1469
```

```r
okcupid[, 1469][1:100] # this  returns the first 100 items of column 1469; this gives an idea of the da
```

```
##   [1] NA                  NA                  NA
##   [4] NA                  NA                  NA
##   [7] NA                  NA                  NA
##  [10] NA                  NA                  NA
##  [13] NA                  NA                  NA
##  [16] "$20,000-$30,000"   NA                  NA
##  [19] NA                  NA                  NA
##  [22] NA                  "$50,000-$60,000"   NA
##  [25] NA                  NA                  NA
##  [28] NA                  NA                  NA
##  [31] "$30,000-$40,000"   "$40,000-$50,000"   NA
##  [34] NA                  NA                  NA
##  [37] NA                  NA                  NA
##  [40] NA                  NA                  NA
##  [43] NA                  "$20,000-$30,000"   "$80,000-$100,000"
##  [46] NA                  "Rather not say"    NA
##  [49] NA                  NA                  NA
##  [52] NA                  NA                  NA
##  [55] NA                  NA                  NA
##  [58] NA                  NA                  "Less than $20,000"
##  [61] NA                  NA                  NA
##  [64] NA                  NA                  NA
##  [67] NA                  NA                  NA
##  [70] NA                  NA                  NA
##  [73] NA                  NA                  NA
##  [76] NA                  NA                  NA
##  [79] NA                  NA                  NA
##  [82] NA                  NA                  NA
##  [85] NA                  "Rather not say"    NA
##  [88] NA                  NA                  "Rather not say"
##  [91] NA                  "$30,000-$40,000"   NA
##  [94] "Rather not say"    NA                  NA
##  [97] NA                  NA                  NA
## [100] NA
```

```r
names(okcupid)[1469] # this returns the name of the 1469th column of okcupid; it is d_income, so you ca
```

```
## [1] "d_income"
```

```r
income <- okcupid[, 1469] # this creates a variable called income that stores the data from all rows of
unique(income) # this returns the unique values; allowing you to see what the range and how variable th
```

```
##  [1] NA                      "$20,000-$30,000"      "$50,000-$60,000"
##  [4] "$30,000-$40,000"       "$40,000-$50,000"      "$80,000-$100,000"
##  [7] "Rather not say"        "Less than $20,000"    "$60,000-$70,000"
## [10] "More than $1,000,000" "$100,000-$150,000"    "$70,000-$80,000"
## [13] "$250,000-$500,000"     "$150,000-$250,000"    "$500,000-$1,000,000"
```

```r
parse_income <- function(x) { # this is creating a new variable that stores a new function
  if (is.na(x) || x == "Rather not say") { # if it has an na or says rather not say
    return(NA_real_) # returns NA
  } else if (x == "More than $1,000,000") { # if the value is "more than 1,000,000
    return(1000000) # return the number 1,000,000
  } else if (x == "Less than $20,000") { # if the value is less than $20,000
    return(20000) # return the number 20,000
  } else if (grepl("-", x)) { # otherwise, see if the elements have -
    nums <- gsub("[\\$,]", "", x) # if they have \\$, it replaces it with a space and stores in num
    parts <- strsplit(nums, "-")[[1]] # this splits the new, stripped down number at the -
    return(mean(as.numeric(parts))) # this returns the mean of the parts as a numeric vector
  }
  return(NA_real_)
}
# this is looking through the income variable and parsing it so it is easier to use and understand. if

okcupid$income_numeric <- as.numeric(sapply(income, parse_income))# a column is added to okcupid that i
# we can then see in the income_numeric column of okcupid there is only 1 value in each row: NA, 1,000,
```

Admittedly, much of the code above is things we have not yet discussed.

Run the code to see what it does. You may want to run individual lines or subsections. Feel free to ask AI specific questions about functions.

Your main job is to add comments to the code to describe what each part does. Furthermore, in your comments indicate the thought process behind the code (i.e. what question prompted the code?).

Putting the code into AI will do this for you - that said, you'll ultimately want to make sure that if I or others gave you the code without comments that you could read the code for yourself.

Finally - make any plot you like to examine the distribution of income and discuss what you observe.

```r
# look at how many unique numbers there are in income to determine range of graph
unique(okcupid$income_numeric)
```

```
##  [1]      NA   25000   55000   35000   45000   90000   20000   65000 1000000
## [10]  125000   75000  375000  200000  750000
```

```r
sort(unique(okcupid$income_numeric))
```

```
##  [1]   20000   25000   35000   45000   55000   65000   75000   90000  125000
## [10]  200000  375000  750000 1000000
```

```r
# there are 14 unique values, including NA, spanning 20,000 - 1,000,000
# there is basically a $10,000 increase with each but it's not steady every increase
table_income_numeric <- (table(okcupid$income_numeric))
table_income_numeric
```

```
##
##   20000   25000   35000   45000   55000   65000   75000   90000  125000  2e+05  375000
##    2819    2480    2241    1805    1359     832     671     891     920    340     121
## 750000   1e+06
##     36     422
```

```
# table to see how many frequencies aree in each answer
# this will help me check my graph is showing what is actually in the variable

# see how many are na
sum_na_income_numeric <- sum(is.na(okcupid$income_numeric))
sum_na_income_numeric
```

```
## [1] 53434
```

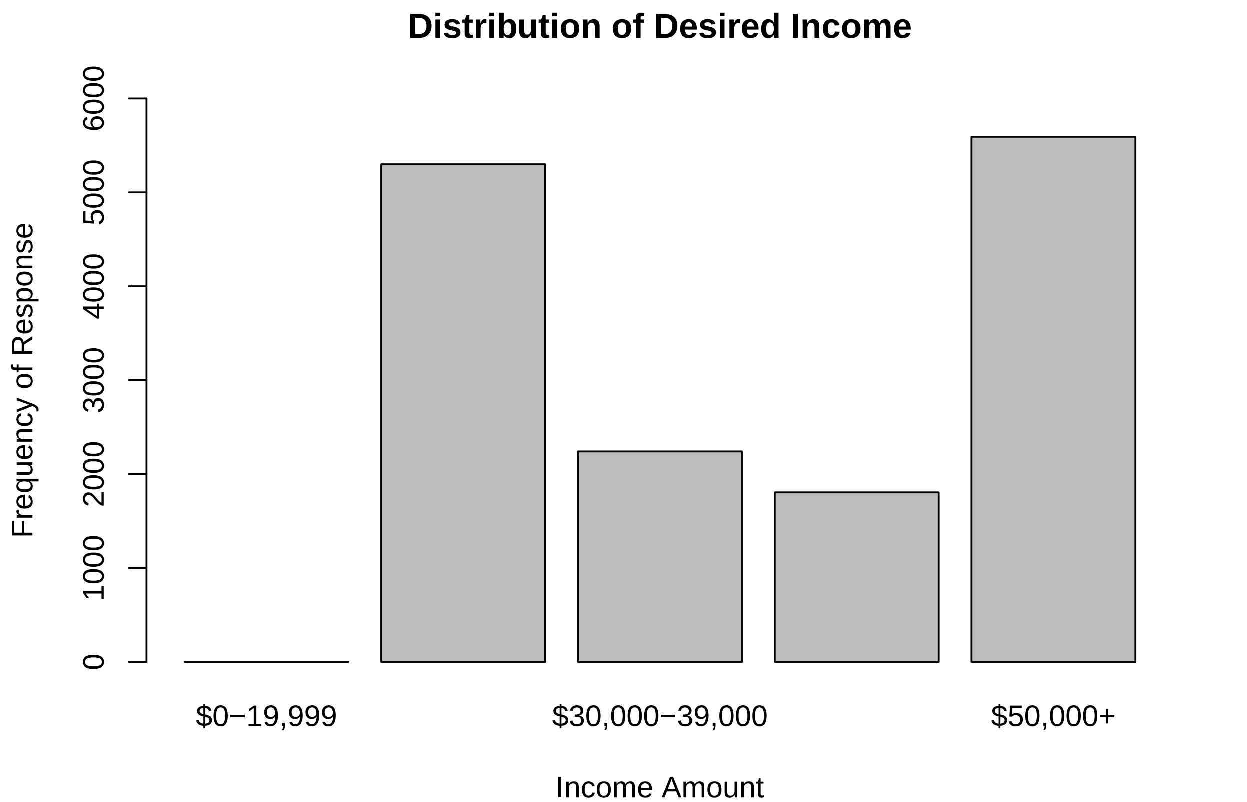
```
# I want to keep the N/As to show how many people did not answer or preferred not to say but 53,000 is
# this will skew th appearance of the rest of the values so I am going to drop them
# A barplot comes to mind but I want the bins to be even step which is better done by a histogram...
# If I make a barplot I need to make specific bins rather than what is the default

# Create a new variable of the bins
income_bins <- cut(
  okcupid$income_numeric,
  breaks = c(0, 20000, 30000, 40000, 50000, 1000000),  # cuts the data at these ranges making 5 bins
  right = FALSE,   # left end included, right end excluded ex: 0-19,999
  include.lowest = TRUE, # very lowest value is included
  labels = c("$0-19,999", "$20,000-29,000", "$30,000-39,000", "$40,000-49,000", "$50,000+") # adds desi
)

# see how many responses are in each level with new binning
table(income_bins)
```

```
## income_bins
##      $0-19,999 $20,000-29,000 $30,000-39,000 $40,000-49,000       $50,000+
##              0           5299           2241           1805           5592
```

```
barplot(table(income_bins),
        main = "Distribution of Desired Income",
        ylab = "Frequency of Response",
        xlab = "Income Amount",
        ylim = c(0,6000))
```



REPLACE THIS TEXT WITH YOUR RESPONSE AND MAKE SURE THE RESPONSE IS IN ITALICS

(1.4) Pick any variable EXCEPT q30458 that starts with `q` that has four possible outcomes (options).

- Create and display a table called `table1` that shows how many of instances there were of each level.
- Write a line that calculates the percentage of values that were from the fourth level of the question, rounded to the nearest whole percent.
- Write a line of code to use paste() to make a statement like "x% of respondents responded (response) to question (text of the question pulled from okcupidQ)."

REPLACE THIS TEXT WITH YOUR R-chunk Code

(1.5) Create a barplot using the `barplot()` function for whatever question you chose in part (1.4).

- The labels of the barplot should also contain the whole number percentages for each possible outcome.
- Format your graph, and make sure the bars are horizontal.
- Change the margins using par(mar = ...) if your labels are being cut off, and change the character size if necessary.

REPLACE THIS TEXT WITH YOUR R-chunk Code

(1.6) Choose any variable that has a Yes/No response.

- Make a two-way table between this new question and your question in 1.4.

- Calculate the percent of people for each level of your 1.4 question that answered 'Yes' to your new question.
- Following the example in Class 4, create a barplot akin to that in lines 431-442 with similar percentages displayed.This will take some adjusting. You can make horizontal or vertical. Make sure no labels are cut off.
- Does it seem like these variables might be related?

REPLACE THIS TEXT WITH YOUR R-chunk Code

REPLACE THIS TEXT WITH YOUR RESPONSE AND MAKE SURE THE RESPONSE IS IN ITALICS

(1.7) Choose any two variables that start with the letter `p`. These are questions that are calculated scales.

- Calculate summary statistics using summary() and also the standard deviaion for each variable (rounded to 2 decimal places).
- Make histograms for each variable. Format your histogram.
- Describe the shape of each histogram (use words like 'symmetric' or 'skewed') and discuss what you observe.

REPLACE THIS TEXT WITH YOUR R-chunk Code

REPLACE THIS TEXT WITH YOUR RESPONSE AND MAKE SURE THE RESPONSE IS IN ITALICS

(1.8) Choose either of the variables you used in 1.7 AND the Yes/No variable you choose in 1.6.

- Get summary statistics for the p_ variable by each level of your Yes/No variable. Compare the medians and means of these two distributions - what do you observe?

REPLACE THIS TEXT WITH YOUR R-chunk Code

REPLACE THIS TEXT WITH YOUR RESPONSE AND MAKE SURE THE RESPONSE IS IN ITALICS

(1.9) Make a scatterplot of your two p_ variables from from 1.7).

- Set a non-default plot character.
- Create the plot using different colors for each level of your binary variable from 1.6.
- Add a legend to your plot
- Format and label your plot
- Discuss any trends you see as well as any differences between groups.

REPLACE THIS TEXT WITH YOUR R-chunk Code

REPLACE THIS TEXT WITH YOUR RESPONSE AND MAKE SURE THE RESPONSE IS IN ITALICS

(1.10) Create a subset of the original dataset that meets the following conditions:

- It includes ONLY the following variables: all five of the variables you have examined so far (income_num, your four level variable, your two p_ indices, and your binary variable) as well as the variable `d_astrology_sign`)
- It has complete data for the variables listed above

- You only want people who have `Ca` or `Sa` as the first two letters of their astrological sign.

Use both the `%in%` operator and the `substr()` function to complete this task. You'll need to look up information about how these work.

Get the dimension of your final dataset. Comment on the reduction in size from the original dataset.

REPLACE THIS TEXT WITH YOUR R-chunk Code

REPLACE THIS TEXT WITH YOUR RESPONSE AND MAKE SURE THE RESPONSE IS IN ITALICS

(1.11) For any one of the questions you wrote in 1.2), investigate this BRIEFLY with some combination of a plot and basic summary statistics (or a table).

REPLACE THIS TEXT WITH YOUR R-chunk Code

REPLACE THIS TEXT WITH YOUR RESPONSE AND MAKE SURE THE RESPONSE IS IN ITALICS

(2.1) Your work at a social research nonprofit puts you in touch with a large, multi-year survey called the National Civic Life Survey. Each row is one respondent; columns include demographics (age, education), civic behaviors (voting frequency, volunteering hours), and attitude questions. Many questions were optional, so missing values are common. The data arrive as a Parquet file, plus a small CSV with question metadata.

Using only the tools you've learned so far, write out in words the steps you would take to explore this dataset and generate a few simple plots. Do not include any R code, but you're welcome to mention R functions you might use.

Hint: In this HW, you worked with parquet/CSV files, inspected structure and summaries, examined unique values for categorical variables, and noted missingness.

REPLACE THIS TEXT WITH YOUR RESPONSE AND MAKE SURE THE RESPONSE IS IN ITALICS